EXPERIMENT 12

Write the Python program for the Tic Tac Toe game

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Aim

To write a Python program to play the Tic Tac Toe game between a human player (X) and the computer (O) on a 3x3 board, where the computer chooses its moves using simple rules.

Algorithm

Step 1: Represent the board as a list of 9 cells (positions 1-9, row by row), all initially empty. The human plays X and moves first.

Step 2: Display the board after every move.

Step 3: Human move: read a position from 1 to 9. If it is outside the range or the cell is occupied, report an invalid move and ask again; otherwise place X there.

Step 4: Computer move: (a) if the computer can win in one move, play it; (b) otherwise, if the human could win in the next move, block it; (c) otherwise take the centre if free; (d) otherwise take a free corner; (e) otherwise take a free side.

Step 5: After every move, check the 8 winning lines (3 rows, 3 columns, 2 diagonals). If any line has the same symbol in all three cells, that player wins and the game ends.

Step 6: If all 9 cells are filled and nobody has won, the game is a draw.

Step 7: Otherwise switch the turn to the other player and repeat from step 3.

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Program

```
import sys

WINS = [(0, 1, 2), (3, 4, 5), (6, 7, 8), (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)]

def show(board):
    for r in range(0, 9, 3):
        print(' ' + ' | '.join(board[r:r + 3]))
        if r < 6:
            print('---+---+---')
    print()

def winner(board):
    for a, b, c in WINS:
        if board[a] != ' ' and board[a] == board[b] == board[c]:
            return board[a]
    return None

def find_winning_move(board, player):
    for pos in range(9):
        if board[pos] == ' ':
            board[pos] = player
            won = winner(board) == player
            board[pos] = ' '
            if won:
                return pos
    return None

def computer_move(board):
    # 1) win  2) block  3) centre  4) corner  5) side
    move = find_winning_move(board, 'O')
    if move is None:
        move = find_winning_move(board, 'X')
    if move is None:
        for pos in (4, 0, 2, 6, 8, 1, 3, 5, 7):
            if board[pos] == ' ':
                move = pos
                break
    return move

def human_move(board):
    while True:
        print("Enter your move (1-9): ", end="", flush=True)
        text = input().strip()
        if not sys.stdin.isatty():
            print(text)                       # echo when input is piped
        if text.isdigit() and 1 <= int(text) <= 9 and board[int(text) - 1] == ' ':
            return int(text) - 1
        print("Invalid move, try again.")

def play():
    board = [' '] * 9
    print("Tic Tac Toe: you are X, computer is O. Positions are numbered 1-9")
    print("row by row (1 = top-left, 9 = bottom-right).\n")
    show(board)
    turn = 'X'
    while True:
        if turn == 'X':
            board[human_move(board)] = 'X'
        else:
            move = computer_move(board)
            board[move] = 'O'
            print(f"Computer plays {move + 1}")
        show(board)
        w = winner(board)
        if w:
            print("You win!" if w == 'X' else "Computer wins!")
```

```
            break
        if ' ' not in board:
            print("It is a draw!")
            break
        turn = 'O' if turn == 'X' else 'X'

if __name__ == "__main__":
    play()
```

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Output

```
Tic Tac Toe: you are X, computer is O. Positions are numbered 1-9
row by row (1 = top-left, 9 = bottom-right).

   |   |
---+---+---
   |   |
---+---+---
   |   |

Enter your move (1-9): 1
 X |   |
---+---+---
   |   |
---+---+---
   |   |

Computer plays 5
 X |   |
---+---+---
   | O |
---+---+---
   |   |

Enter your move (1-9): 5
Invalid move, try again.
Enter your move (1-9): 9
 X |   |
---+---+---
   | O |
---+---+---
   |   | X

Computer plays 3
 X |   | O
---+---+---
   | O |
---+---+---
   |   | X

Enter your move (1-9): 7
 X |   | O
---+---+---
   | O |
---+---+---
 X |   | X

Computer plays 4
 X |   | O
---+---+---
 O | O |
---+---+---
 X |   | X

Enter your move (1-9): 6
 X |   | O
---+---+---
 O | O | X
---+---+---
 X |   | X

Computer plays 8
 X |   | O
---+---+---
 O | O | X
---+---+---
 X | O | X
Enter your move (1-9): 2
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```
 X | X | O
---+---+---
 O | O | X
---+---+---
 X | O | X

It is a draw!
```

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